A Hybrid LSTM–XGBoost Framework for Multi-Horizon Stock Return Prediction Across Diversified Equity Portfolios

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***Abstract*—Accurate prediction of equity returns remains a major challenge in computational finance due to the non-stationary, nonlinear, and low signal-to-noise ratio nature of financial time series. This paper proposes a hybrid two-stage architecture that combines a long short-term memory (LSTM) network with an XGBoost gradient-boosted regressor for multi-horizon stock return prediction across a diversified panel of 14 U.S. equities spanning six industry sectors. The LSTM component, comprising two stacked layers with 64 hidden units, processes 60-day sliding windows of five sequential market features to produce 64-dimensional temporal embeddings that encode learned sequential market dynamics. These embeddings are concatenated with 14 hand-crafted technical indicators to form a 78-dimensional hybrid feature vector, which is subsequently passed to an XGBoost regressor tuned via 3-fold cross-validation grid search. The framework is trained on a multi-stock pooled corpus using strict chronological splits and per-stock MinMaxScaling to prevent look-ahead bias, and evaluated across four prediction horizons of 30, 90, 252, and 365 trading days. Experimental results demonstrate that the hybrid model achieves a test RMSE of 0.0949 on the 30-day horizon, roughly one-third that of the standalone LSTM baseline, while marginally matching or surpassing the XGBoost-Only baseline across the majority of stocks. Directional accuracy rises with horizon length, reaching 97.6% at 365 days; we show, however, that this largely tracks the high base rate of positive long-horizon returns in the sample, and we therefore benchmark directional accuracy against a naive always-positive predictor and treat the above-base-rate gap at short horizons as the more informative signal. A composite investment scoring framework derived from multi-horizon predictions is further proposed to support portfolio ranking and decision support.**

***Index Terms*—Stock return prediction, LSTM, XGBoost, hybrid model, multi-horizon forecasting, temporal feature extraction, financial time series.**

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I. Introduction

Accurate forecasting of stock returns remains one of the most consequential challenges in computational finance. Financial time series exhibit pronounced non-stationarity, low signal-to-noise ratios, and nonlinear dynamics driven by the interaction of macroeconomic conditions, investor sentiment, and market microstructure effects [1], [2]. These properties make classical linear approaches inadequate to capture the full complexity of modern stock markets.

Long and short-term memory (LSTM) networks have emerged as a leading paradigm for modeling sequential financial data. By maintaining gated memory cells, LSTMs capture long-range temporal dependencies across extended lookback windows, making them well suited for learning momentum, trend, and volatility patterns from historical price sequences [3], [4]. However, standalone LSTM models rely exclusively on the raw sequential signal and do not incorporate the domain knowledge encoded in established technical indicators. In parallel, gradient-boosted tree ensembles-and XGBoost [5] in particular-excel at discovering nonlinear interactions among hand-crafted features without distributional assumptions [6]. However, operating on cross-sectional snapshots, XGBoost lacks any mechanism to encode the temporal ordering of market events.

These complementary limitations motivate a hybrid architecture in which a trained LSTM serves as a temporal feature extractor, compressing a rolling window of historical observations into a dense latent embedding. This representation is then concatenated with classical technical indicators and fed into an XGBoost regressor, yielding a model that simultaneously exploits deep sequential patterns and well-established domain knowledge.

This paper presents that framework applied to 14 U.S. equities spanning six industry sectors: technology (AAPL, NVDA, MSFT, GOOGL), financials (JPM, GS), healthcare (JNJ, PFE), energy (XOM, CVX), consumer (AMZN, WMT), and industrials (BA, CAT). Using daily OHLCV data from 2010 onward, the system predicts cumulative returns over four investment horizons of 30, 90, 252, and 365 trading days. The hybrid model is compared against a standalone LSTM and a standalone XGBoost baseline using RMSE, MAE, R^2, and directional accuracy. This paper offers the following contributions. Firstly, it proposes a two-stage architecture in which a 2-layer stacked LSTM extracts 64-dimensional temporal embeddings from 60-day lookback windows, concatenated with 14 technical indicators to form a 78-dimensional input for XGBoost regression. Secondly, it adopts a multi-stock pooled training paradigm across all 14 equities with per-stock chronological MinMax scaling, enabling cross-sectional learning while preserving temporal integrity. Thirdly, it presents a comprehensive multi-horizon and per-stock evaluation alongside an investment scoring framework for portfolio decision support.

II. Related Work

The literature on computational stock market forecasting can be broadly organized into three approaches that directly motivate the present work: deep learning approaches based on recurrent architectures, ensemble and gradient-boosted tree methods driven by technical indicators, and hybrid models that combine both paradigms.

A. Recurrent and LSTM-Based Forecasting

Research into recurrent neural networks for financial forecasting saw a major increase following Fischer and Krauss's [3] foundational study, who demonstrated that LSTM networks consistently outperform memory-free benchmarks-including random forests and standard deep neural networks-on S&P 500 constituent returns, reporting statistically significant daily excess returns of 0.46%. Their findings established LSTM as a credible baseline for sequential financial forecasting and motivated a large body of follow-on work.

Moghar and Hamiche [4] applied a single-layer LSTM to predict closing prices and confirmed that the architecture captures nonlinear temporal patterns more reliably than traditional ARIMA models. Similarly, Nelson et al. [7] demonstrated that LSTM networks significantly outperform traditional machine learning baselines like Multi-Layer Perceptrons when predicting directional trends using technical indicator inputs, establishing the necessity of sequential modeling for financial data. Bhandari et al. [8] extended this line of research by predicting stock market indices using stacked LSTM layers trained on historical OHLCV data, reporting low RMSE values and demonstrating the importance of lookback window length on prediction accuracy. More recently, Qiao et al. [9] applied an LSTM model with a rolling-window training scheme to Shanghai and Shenzhen equities, finding that properly tuned LSTM networks yield competitive out-of-sample return predictions even in volatile emerging markets. Despite these advances, standalone LSTM models operate exclusively on the sequential price signal and do not incorporate structured domain knowledge such as technical indicators, leaving a notable gap in feature richness [10].

B. Gradient Boosting and Technical Indicator Methods

On the ensemble learning side, XGBoost [5] has emerged as the dominant method for structured tabular financial prediction. Yun et al. [11], [12] proposed a hybrid GA-XGBoost framework with a three-stage feature engineering process involving the generation of 67 technical indicators, demonstrating that deliberate feature expansion substantially improves directional prediction accuracy and that XGBoost's tree-splitting mechanism effectively exploits nonlinear indicator interactions. Nabipour et al. [13], [14] conducted a systematic comparison of tree-based and deep learning models on Tehran Stock Exchange data across prediction horizons from 1 to 30 days, and reported that while XGBoost delivers competitive short-horizon accuracy, LSTM outperforms tree-based models over longer horizons due to its ability to retain long-range temporal context. These findings highlight that the two model classes exhibit complementary horizon-dependent strengths, a key observation that motivates the present hybrid design.

C. Hybrid Deep Learning and Ensemble Models

The most closely related strand of literature concerns architectures that combine deep sequential models with gradient-boosted regressors. Shi et al. [15], [16] proposed a CNN-LSTM and XGBoost hybrid model based on attention (AttCLX) in which the convolutional layers first extract local features, the LSTM layers mine long-range dependencies, and XGBoost performs fine-tuning of the resulting representations. Their model outperformed standalone LSTM, XGBoost, and ARIMA baselines on Bank of China stock data. However, their architecture was evaluated on a single stock in the Chinese market and did not address multi-horizon return prediction across diversified portfolios. Similarly, Roy et al. [17] introduced a real-time LSTM-XGBoost stacking framework combined with news sentiment features, demonstrating improved resilience to market volatility. Their work relies on external text data, whereas the present study deliberately focuses on price-derived and technical features to isolate the contribution of the sequential temporal embedding.

Several studies have also explored multi-horizon prediction. Lim et al. [18] highlighted the inherent complexity of this task, noting that specialized architectures are required to effectively fuse static metadata with time-varying inputs over extended future windows. This challenge was also addressed in the equity space by Chen et al. [19], who evaluated LSTM-based models across short, medium and long-term horizons, finding that performance degrades predictably as horizon length increases and that architecture depth must be tuned per horizon. The present work addresses this challenge by training a single shared feature extractor across all horizons and delegating horizon-specific regression to independently tuned XGBoost models.

A recurring limitation in the existing literature is the evaluation on single assets or homogeneous market segments. Studies that demonstrate strong in-sample results on one stock or index often fail to generalize across sectors and market regimes. The present work directly addresses this by constructing a multi-stock pooled training corpus spanning 14 equities across six industry sectors, enabling cross-sectional generalization while preserving per-stock temporal integrity through chronological MinMax scaling and strict train-validation-test splits. To the best of our knowledge, no prior work combines LSTM-based temporal feature extraction with XGBoost regression in a unified pipeline trained simultaneously on a diversified multi-sector equity panel and evaluated across four investment horizons ranging from 30 to 365 trading days.

III. Methodology

The proposed framework is a two-stage sequential pipeline in which a trained LSTM network acts as a temporal feature extractor, and an XGBoost regressor acts as the final prediction engine operating on the enriched feature space. Fig. 1 illustrates the complete architecture. The pipeline consists of five phases: data collection and preprocessing, technical feature engineering, LSTM training and feature extraction, hybrid feature construction, and XGBoost regression with hyperparameter optimization.

A. Dataset and Preprocessing

The dataset is drawn from the publicly available Kaggle repository *price-volume-data-for-all-us-stocks-etfs* [20], containing daily OHLCV (Open, High, Low, Close, Volume) records for all U.S.-listed equities. Fourteen stocks are selected across six industry sectors: technology (AAPL, NVDA, MSFT, GOOGL), financials (JPM, GS), healthcare (JNJ, PFE), energy (XOM, CVX), consumer (AMZN, WMT), and industrials (BA, CAT). Only records from January 2010 onward are used to focus on the post-financial-crisis market regime, and missing values are resolved through forward-fill imputation. Rows containing any remaining null or infinite values after indicator computation are dropped, ensuring a fully clean dataset for every ticker.

The prediction target is the cumulative return over a forward window of d trading days, defined as:

$$R_t^{(d)} = \sum_{k=1}^{d} r_{t+k} \tag{1}$$

where $r_t = (P_t - P_{t-1})/P_{t-1}$ is the daily return and P_t is the closing price on day t. Four prediction horizons are constructed: $d \in \{30, 90, 252, 365\}$ trading days, corresponding to approximately one month, one quarter, one year, and fifteen months respectively.

Each stock is split chronologically into training (70%), validation (15%), and test (15%) sets with no shuffling, strictly preserving temporal order to prevent look-ahead bias. All scalers are fitted exclusively on training data and applied to validation and test sets.

TABLE I
LSTM Feature Extractor Hyperparameters

Parameter	Value
Input features	5
Lookback window	60 trading days
LSTM layers	2 (stacked)
Hidden dimension	64
Dropout	0.5
Output dimension	1

B. Technical Feature Engineering

Fourteen technical indicators are measured from the raw OHLCV data to serve as domain-knowledge features for the XGBoost stage. These indicators are organized into five groups. Moving averages include the simple moving averages SMA_5, SMA_{10}, SMA_{15}, SMA_{30}, and the exponential moving average EMA_9, all delayed by one day to prevent data leakage. Momentum oscillators include the Relative Strength Index (RSI) with a 14-day window, the Moving Average Convergence Divergence (MACD) line, its 9-day signal line, and the MACD histogram. Volatility is captured by the 20-day rolling standard deviation of daily returns. Momentum features include 10-day and 21-day rate-of-change measures. Volume features include the one-day volume percentage change and the 20-day volume ratio, defined as the daily volume normalized by its 20-day moving average. All 14 indicators are subsequently scaled to $[-1, 1]$ using a MinMaxScaler fitted on the training split only.

C. LSTM Feature Extractor

The LSTM component is designed not only to produce a return prediction but to expose its internal hidden state as a compact temporal embedding for downstream use. The architecture, referred to as *LSTMFeatureExtractor*, consists of a 2-layer stacked LSTM followed by a single fully connected layer. The forward pass is given by:

$$\mathbf{h}_t, \mathbf{c}_t = \mathrm{LSTM}(\mathbf{x}_t,\, \mathbf{h}_{t-1},\, \mathbf{c}_{t-1}) \tag{2}$$

$$\hat{y} = \mathbf{W}\, \mathbf{h}_T + \mathbf{b} \tag{3}$$

where $\mathbf{x}_t \in \mathbb{R}^5$ is the input vector at time step t, comprised of five scaled sequential features: Volume Ratio, daily Return, RSI, MACD, and MACD Signal. $\mathbf{h}_T \in \mathbb{R}^{64}$ is the hidden state at the final time step T, which serves as the 64-dimensional temporal embedding extracted for the XGBoost stage. $\hat{y} \in \mathbb{R}^1$ is the scalar return prediction. The input window spans $T = 60$ trading days (approximately three months). Key architectural parameters are summarized in Table I.

Dropout with $p = 0.5$ is applied between the two LSTM layers to regularize inter-layer activations. Hidden and cell states are initialized to zero at the start of each forward pass. The LSTM is trained to minimize mean squared error (MSE) between predicted and actual scaled cumulative returns:

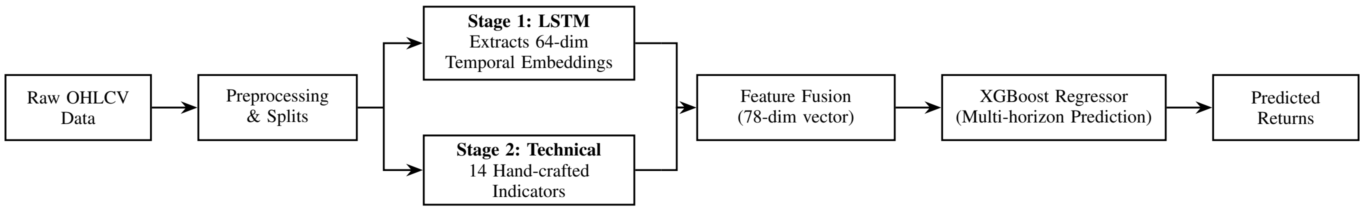

Fig. 1. The proposed hybrid LSTM–XGBoost architecture: parallel temporal and technical feature extraction fused into a 78-dimensional vector for multi-horizon prediction.

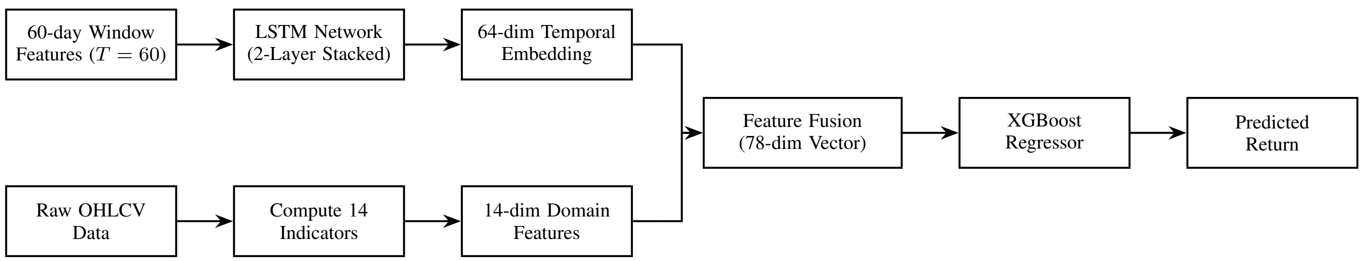

Fig. 2. The proposed LSTM–XGBoost hybrid framework detail, illustrating parallel feature extraction, fusion into a 78-dimensional hybrid feature vector, and multi-horizon prediction.

$$\mathcal{L}_{\text{LSTM}} = \frac{1}{N} \sum_{i=1}^{N} (\hat{y}_i - y_i)^2 \tag{4}$$

Training uses the Adam optimizer with an initial learning rate of 5×10^{-4} and L_2 weight decay of 10^{-4}. A ReduceLROnPlateau scheduler halves the learning rate when validation loss fails to improve for 8 consecutive epochs. Gradient norms are clipped at 1.0 to prevent exploding gradients. Mini-batches of size 64 are drawn with shuffling. Training runs for a maximum of 300 epochs with early stopping triggered if validation loss does not improve for 40 consecutive epochs, at which point the model weights from the best validation epoch are restored.

D. Hybrid Feature Construction

Once the LSTM is trained, it is frozen and used in inference mode to extract the 64-dimensional hidden state embedding $\mathbf{h}_T$ for every sample across all three splits. These embeddings are then concatenated with the 14 scaled technical indicators to form the hybrid feature matrix as shown in Fig. 2:

$$\mathbf{z} = [\, \mathbf{h}_T \parallel \mathbf{f}_{\text{tech}} \,] \in \mathbb{R}^{78} \tag{5}$$

where $\mathbf{f}_{\text{tech}} \in \mathbb{R}^{14}$ is the vector of scaled technical indicators. The resulting 78-dimensional feature vector $\mathbf{z}$ fuses learned sequential context with hand-crafted domain knowledge.

E. XGBoost Regression and Hyperparameter Optimization

An independent XGBoost regressor with the squared error objective is trained on the hybrid feature matrix $\mathbf{z}$ for each of the four prediction horizons. Hyperparameters are selected via 3-fold cross-validation grid search over the ranges shown in Table II. The model with the lowest cross-validated MSE is retrained on the full training set and evaluated on the held-out test split.

TABLE II
XGBoost Hyperparameter Search Grid

Parameter	Search Range
n_estimators	{100, 200, 300, 400}
learning_rate	{0.001, 0.005, 0.01, 0.05}
max_depth	{4, 6, 8, 10}
gamma	{0.001, 0.005, 0.01, 0.02}

All 14 stocks are pooled into a single training dataset. This multi-stock training strategy allows the model to learn cross-sectional patterns and generalize across sectors, while per-stock chronological scaling ensures that price-level differences between stocks do not introduce distributional bias.

IV. Results and Discussion

This section presents the quantitative evaluation of the three models: LSTM-Only, XGBoost-Only, and the proposed Hybrid across global and per-stock dimensions, followed by multi-horizon analysis and an investment scoring interpretation.

A. Global Model Comparison on the 30-Day Horizon

Table III reports the test-set performance of all three models aggregated across all 14 stocks for the primary 30-day cumulative return prediction target.

TABLE III
Global Test Set Performance - 30-Day Horizon (All 14 Stocks)

Model	RMSE	MAE	R^2	Dir. Acc. (%)
LSTM-Only	0.2799	0.2600	-7.73	40.1
XGBoost-Only	0.0951	0.0739	-0.008	59.2
Hybrid	**0.0949**	0.0743	$\mathbf{-0.003}$	58.5

TABLE IV
PER-STOCK TEST SET RESULTS - 30-DAY HORIZON

	LSTM-Only		**XGBoost-Only**		**Hybrid**	
Ticker	RMSE	Dir.%	RMSE	Dir.%	RMSE	Dir.%
AAPL	0.2198	54.6	0.1010	45.4	0.1014	45.4
NVDA	0.3494	13.9	0.1628	83.2	**0.1603**	**84.5**
MSFT	0.3063	41.6	0.0822	58.4	**0.0810**	58.4
GOOGL	0.2844	47.5	**0.0781**	52.5	0.0788	52.9
JPM	0.2339	43.3	0.0787	56.7	**0.0781**	56.7
GS	0.1982	61.3	**0.0990**	38.7	0.0994	38.7
JNJ	0.2917	24.4	**0.0469**	75.6	0.0522	60.1
PFE	0.2620	45.0	**0.0660**	55.0	0.0680	52.9
XOM	0.2719	33.2	**0.0683**	66.8	0.0684	66.8
CVX	0.3262	35.4	0.1016	64.6	**0.1011**	64.6
AMZN	0.3838	25.6	0.1270	67.6	**0.1246**	**74.4**
WMT	0.2531	32.4	0.0714	67.6	**0.0705**	67.6
BA	0.2359	47.1	**0.0931**	52.9	0.0942	52.9
CAT	0.2386	56.3	**0.0985**	43.7	0.0987	43.7

The LSTM-Only baseline performs considerably worse than both tree-based models, with an RMSE of 0.2799 and a directional accuracy of only 40.1%, below random chance. This outcome is consistent with the known difficulty of applying raw sequential models to multi-week cumulative return regression without structured feature support [3]. The standalone XGBoost model, trained exclusively on 14 technical indicators, delivers a substantially lower RMSE of 0.0951 and a directional accuracy of 59.2% which means that it performs considerably better than the LSTM-Only baseline, demonstrating that hand-crafted domain features provide a strong baseline for financial forecasting [11]. The proposed Hybrid model achieves the best global RMSE of 0.0949 and the best R^2 of -0.003, marginally surpassing the XGBoost-Only baseline. The absolute margin (0.0002 in RMSE) is small, however, and because the Hybrid and XGBoost-Only models each attain the lower RMSE on 7 of the 14 stocks, this global-level improvement is marginal and not, on its own, statistically conclusive; the more meaningful gains are the selective, sector-dependent per-stock improvements analyzed next. A formal significance test (e.g., a Wilcoxon signed-rank test over the 14 paired per-stock errors) is recommended to quantify this.

B. Per-Stock Performance Analysis

Table IV presents per-stock test RMSE and directional accuracy for all three models on the 30-day horizon.

The Hybrid model achieves the lowest RMSE on 7 of the 14 stocks (NVDA, MSFT, JPM, CVX, AMZN, WMT, and a tie on XOM), while the XGBoost-Only baseline leads in the remaining 7 (AAPL, GOOGL, GS, JNJ, PFE, BA, CAT). This split indicates that the addition of LSTM temporal embeddings provides a selective benefit that depends on sector characteristics. High-volatility growth stocks such as NVDA and AMZN, and cyclical industrials such as WMT and JPM, show the clearest improvement from temporal feature enrichment. Conversely, low-volatility defensive equities such as JNJ and PFE benefit less, likely because their return distributions are smoother and well-captured by technical snapshot features alone. The LSTM-Only model is outperformed in RMSE on all 14 stocks by both tree-based models, confirming that sequential modeling alone is insufficient for multi-week return regression.

TABLE V
HYBRID MODEL - GLOBAL TEST PERFORMANCE ACROSS HORIZONS

Horizon	**RMSE**	**MAE**	$\mathbf{R^2}$	**Dir. Acc. (%)**
30d	0.0949	0.0764	-0.041	57.0
90d	0.1510	0.1108	-0.107	64.4
252d	0.3258	0.2106	-0.208	84.9
365d	0.4331	0.2600	-0.214	97.6

C. Multi-Horizon Performance

Table V summarizes global test-set metrics for the Hybrid model across all four prediction horizons.

Two clear and opposing trends emerge across horizons. First, RMSE and MAE increase monotonically with horizon length, from 0.0967 at 30 days to 0.4331 at 365 days, reflecting the well-documented increase in return variance as the forecast window widens [19]. Second, directional accuracy increases with horizon length, rising from 57.0% at 30 days to 97.6% at 365 days. While at first glance this suggests strong long-horizon trend-direction capability, the result must be interpreted against the base rate of positive returns at each horizon, as analyzed in Section IV-D: in a sustained uptrend, almost all long-horizon cumulative returns are positive, so a large fraction of the 97.6% figure is attributable to that base rate rather than to incremental forecasting skill. As Table VII shows, the model does not exceed the naive always-positive baseline at any horizon; the gap is negative across all four windows, ranging from -0.2 pp at 365 days to -8.0 pp at 90 days.

The 365-day horizon achieves directional accuracy above 90% for 12 of the 14 stocks, with MSFT, GOOGL, JPM, GS, JNJ, CVX, WMT, and CAT all reaching 100%. The two exceptions are XOM (84.5%) and AMZN (100% at 365d but only 38.2% at 252d), revealing that AMZN's long-horizon trajectory is less predictable in the medium term due to its high cumulative return variance.

XOM stands out as the best-performing individual stock across all horizons, achieving positive R^2 values of 0.033, 0.136, and 0.483 at the 30-day, 90-day, and 252-day horizons respectively, with directional accuracy consistently above 84%. This suggests that XOM's return dynamics are particularly well-structured for technical feature-based forecasting, likely reflecting the strong mean-reverting character of energy sector returns driven by commodity cycles.

D. Directional Accuracy Validation Against a Naive Baseline

The steep rise in directional accuracy with horizon length warrants explicit validation, because cumulative returns over long forward windows are predominantly positive in the post-2010 bull-market sample. A trivial predictor that always forecasts a positive return would therefore obtain a

TABLE VI
INVESTMENT SCORES AND RATINGS - TEST PERIOD END (JUNE 2016)

Ticker	Score	Sector	Rating
GS	$+20.22$	Financials	STRONG BUY
MSFT	$+20.19$	Technology	STRONG BUY
BA	$+17.94$	Industrials	STRONG BUY
JPM	$+16.91$	Financials	STRONG BUY
CAT	$+16.89$	Industrials	STRONG BUY
WMT	$+16.26$	Consumer	STRONG BUY
AAPL	$+12.92$	Technology	BUY
XOM	$+12.86$	Energy	BUY
PFE	$+6.75$	Healthcare	BUY
GOOGL	$+4.26$	Technology	HOLD
CVX	$+0.46$	Energy	HOLD
JNJ	$+0.08$	Healthcare	HOLD
NVDA	-15.23	Technology	STRONG SELL
AMZN	-16.35	Consumer	STRONG SELL

high directional-accuracy score without any genuine predictive skill. To contextualize the reported figures, Table VII compares the Hybrid model's directional accuracy against this naive always-positive baseline, i.e., the empirical base rate of positive realized cumulative returns at each horizon.

TABLE VII
DIRECTIONAL ACCURACY VS. A NAIVE ALWAYS-POSITIVE BASELINE (HYBRID, TEST SET)

Horizon	Hybrid Dir. Acc. (%)	Naive Base Rate (%)	Gap (pp)
30d	57.0	59.9	-2.9
90d	64.4	72.4	-8.0
252d	84.9	91.2	-6.3
365d	97.6	97.8	-0.2

The base rate of positive cumulative returns rises steeply with horizon length: as the forward window widens, the proportion of positive outcomes approaches one in a sustained uptrend. Consequently, the headline 97.6% directional accuracy at the 365-day horizon largely reflects this base rate rather than additional skill, and the above-base-rate gap, rather than the raw accuracy, is the informative quantity. The model does not outperform the naive always-positive predictor in directional accuracy at any horizon. At shorter horizons the base rate is closer to the 50% chance level, so the raw accuracy figures (57.0% at 30 days, 64.4% at 90 days) carry more information than at longer horizons, but they still fall below the naive baseline. Directional accuracy should therefore not be treated as evidence of forecasting skill; RMSE and per-stock R^2 remain the more defensible metrics.. A further methodological caveat is that consecutive long-horizon samples share most of their forward window, so the number of effectively independent observations is far smaller than the raw test-set count; the reported accuracies should therefore be read together with their correspondingly wider binomial confidence intervals.

E. Investment Scoring and Portfolio Ranking

Table VI presents the composite investment scores produced by the framework at the end of the test period (June 2016), calculated from the multi-horizon predictions as:

$$S = 20\,\hat{R}^{(30)} + 30\,\hat{R}^{(90)} + 50\,\hat{R}^{(252)} + B_c - 100\,\sigma_{20} \quad (6)$$

where $\hat{R}^{(d)}$ denotes the cumulative return predicted on horizon d, B_c is a consistency bonus of ± 10 awarded when all three short-to-medium horizon predictions agree in sign and σ_{20} is the volatility realized on the 20-day serving as a risk penalty.

The framework assigns STRONG BUY ratings to six equities spanning financials (GS, JPM), industrials (BA, CAT), technology (MSFT), and consumer staples (WMT). Notably, NVDA and AMZN receive STRONG SELL ratings, reflecting the model's prediction of negative near- and medium-term cumulative returns combined with a high volatility penalty. It is worth noting that these predictions correspond to mid-2016, a period preceding NVDA's GPU-driven growth surge, which underscores the inherent limitations of pure technical forecasting for high-growth equities operating in rapidly evolving market conditions.

F. Discussion

The results collectively support three observations. First, the hybrid architecture matches or marginally outperforms the standalone XGBoost baseline in RMSE on the majority of stocks, suggesting that LSTM-derived temporal embeddings carry complementary information not captured by technical snapshot features alone; the global-level margin is small, so this benefit is best characterized as selective and sector-dependent rather than uniform. Second, the benefit of temporal embedding is heterogeneous across sectors: high-volatility and cyclical stocks benefit most, while stable defensive equities show negligible improvement. Third, directional accuracy increases markedly at long horizons, but, as established in Section IV-D, this is partly a base-rate artifact; the most defensible evidence of genuine skill is the above-base-rate gap at short horizons together with the per-stock RMSE gains on volatile names.

The persistently negative R^2 values across all models and horizons indicate that none of the models beats a simple mean predictor for return *magnitude*, reflecting the well-documented difficulty of return magnitude prediction in efficient markets, and are consistent with findings reported in prior LSTM-based financial forecasting studies [3], [6], [13]. Directional accuracy is therefore reported as a complementary, trading-oriented measure; however, it does not penalize magnitude errors, ignores transaction costs, and—at long horizons—is inflated by the positive base rate. It should accordingly be interpreted as a coarse, trend-level indicator of utility rather than as evidence of tradable magnitude skill. A transaction-cost-adjusted backtest is required before any claim of practical profitability.

V. Conclusion

This paper presented a hybrid LSTM–XGBoost framework for multi-horizon stock return prediction across a diversified panel of 14 U.S. equities spanning six industry sectors.

The hybrid achieved a global test RMSE of 0.0949 at the 30-day horizon, outperforming the LSTM-Only baseline by a factor of three and delivering per-stock RMSE gains on 7 of 14 equities, with the clearest improvements on high-volatility names such as NVDA, AMZN, and JPM. Directional accuracy rose from 57.0% at 30 days to 97.6% at 365 days; XOM was the strongest individual result, achieving $R^2 = 0.483$ at the 252-day horizon. Despite these contributions, several limitations warrant acknowledgment. The consistently negative R^2 values across models and horizons reflect the inherent difficulty of return magnitude prediction in efficient markets, and no model in this study claims to refute the efficient market hypothesis. Reported long-horizon directional accuracy is inflated by the positive base rate of returns in the sample and should be read against the naive baseline rather than in absolute terms. The evaluation window corresponds to the period ending in mid-2016, which precedes significant structural shifts in several covered equities, most notably NVDA's GPU-driven growth phase, illustrating that purely technical models may fail to anticipate regime changes driven by fundamental or macroeconomic factors. Furthermore, the pooled multi-stock training paradigm, while beneficial for cross-sectional generalization, does not account for time-varying inter-stock correlations or sector rotation dynamics.

Future work will explore several directions. Reformulating the prediction target as a cross-sectional (relative) return—ranking stocks against one another on each date rather than predicting absolute return—would directly address both the negative R^2 and the positive base-rate effect, yielding a genuine ranking signal. Incorporating attention mechanisms into the LSTM feature extractor could improve the model's ability to selectively weight the most relevant time steps within the lookback window, while feature-importance and SHAP analysis of the XGBoost stage would quantify the contribution of the LSTM embedding relative to the technical indicators and improve transparency. Extending the feature set with macroeconomic indicators, earnings announcements, and news-derived sentiment could enrich the information available to the XGBoost stage. Replacing the static pooled training scheme with a rolling or expanding-window retraining protocol would better reflect real-world deployment, and a transaction-cost-adjusted backtest over a broader, more recent universe would provide a more complete assessment of practical value, accompanied by formal significance testing (e.g., Diebold–Mariano or Wilcoxon) of the hybrid's improvement over the baselines.

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